

Element F

Consideration of design viability

The proposed design was carefully reviewed based on several relevant extra-functional considerations; a judgment about design viability based on those considerations — the capacity of the proposed solution to address the problem — is clearly realistic and well supported with credible evidence.

Throughout many points in this journey, our group has gone through several specific designs and iterations of designs before reaching our final design. The Andean Bear Zookeepers challenged with the following criteria they wished to incorporate into our design.

A. Durability: Materials used should be sturdy and resilient to withstand the strength of the bears. PVC (schedule 40 or schedule 80) and thick wood (4x4) are recommended.

B. Ease of Cleaning: Materials should be easy to clean to maintain hygiene standards.

Temporary Design: The puzzle should be temporary to allow for movement within the habitat and dens.

C. Feeding-based Enrichment: The puzzle should primarily revolve around food to incentivize the bears. However, other interactive elements like items to push can be included.

D. Customization: The puzzle should be customizable to accommodate different sizes of food items, with the largest portion being 2"x2" to 3".

E. Dimensions: The maximum size of the puzzle feeder should not exceed the dimensions of a 31-gallon water drum (22 inches in diameter, 35 inches tall).

- F. Mounting Requirement: Feeders for the habitat must be mounted securely to prevent them from falling into the moat, pool, or hitting the glass. Inside dens, options for both mounted and unmounted feeders are acceptable.
- G. Attention Span Consideration: The puzzle should engage the bears for as long as possible, ideally allowing them to return to it throughout the day. Attention spans vary but can range from 5-10 minutes or until food is depleted.

For criteria A, we opted to use high-molecular-weight high-density polyethylene (HMW-HDPE) than that of the recommended schedule 40 PVC. Although PVC has a higher tensile strength than HMW-HDPE, the processing or cutting of PVC can release chloride gas, compromising the safety of both bear consumption and student-use. HMW-HDPE is also more resistant to fatigue cracking and failure due to its high concentration of polyethylene and used for food-grade applications. See the resources page for more in-depth information.

For criteria B, the group decided to make disassembly an easy task. This allowed the final design to be a simplification of the previous iterations and improved for optimal use. However, some aspects of the final product deemed to be harder to disassemble in person than in theory. This may be due to human error when manufacturing the puzzle. Additionally, some parts of the barrel may not be washed due to the potential formation of pollutants or malfunction of the puzzle. For example, the lazy susan may not be washed as the grease to allow the lazy susan will wear off. Additionally, washing the grease off onto certain surfaces can damage the chemical integrity of plastics. This can be solved by avoiding washing the lazy susan, which disassembly can prove to be beneficial.

For criteria C, the use of a large container allows for room for zookeepers to add as much for attracting the bear's attention through gaps in the lid that allow for the scent of the food to pass through, but not necessarily the food itself.

For criteria D, the customization of the puzzle can be varied as smaller objects can be included inside of the puzzle to add combinations and variables to the puzzle. This can be altered freely by the zookeepers as the large barrel can accommodate fairly mid-sized items.

For criteria E, the dimensions of the project is that of the listed size.

For criteria F, the outer parts of the puzzle consist of a clamp that seals the lid of the barrel to the lip of the barrel. This clamp also has a pass-through hole to lock the clamp in place, allowing the zookeepers to secure the lid close from the bear. This lock (sold separately) can be used to attach a chain to the top the barrel, securing the barrel to the enclosure. The puzzle is not required to have an attachment point, but having the option to do so can provide the randomness factor and customization ability to the list.

For criteria G, the design incorporates randomness as a factor to allow the bear to face the same object, but not the same objective. This was done by the addition of a human-removable handle that will then provide the factor of randomness by allowing room for the bear to make errors when trying to repeat the task of opening the lid the same fashion it has done before. This ensures that the bear will never come across the same exact puzzle, but rather a variation.

Resources:

PVC vs. HDPE

[https://wlplastics.com/hdpe-vs-pvc-pipes/#:~:text=Polyvinyl%20chloride%20\(PVC\)%20has%20a,to%20fatigue%20cracking%20and%20failures.](https://wlplastics.com/hdpe-vs-pvc-pipes/#:~:text=Polyvinyl%20chloride%20(PVC)%20has%20a,to%20fatigue%20cracking%20and%20failures.)

HDPE daily applications

<https://www.shell.us/business-customers/shell-polymers/resources-and-insights/polyethylene-in-everyday-life-common-hdpe-plastic-products.htm>